Problem Identification

Problem Statement

Many people face barriers when trying to engage with new domains such as sports, politics, finance, or nutrition. News articles and commentary often assume a baseline of prior knowledge and use domain-specific jargon, making content inaccessible to beginners.

This creates an “accessibility gap” in information, discouraging users from exploring new interests or staying informed. Additionally, individuals with disabilities encounter further barriers, as news platforms frequently lack robust accessibility support (i.e. screen readers, adjustable text size, high-contrast modes). The result is that many readers are excluded from meaningful participation in news, current events, and social conversations.

Initial Project Proposal

The News for U platform will address these challenges by creating an accessible, AI-powered news aggregator that adapts to the user’s level of familiarity with a topic. By integrating large language models (LLMs) and accessibility features, it will simplify news articles, explain jargon, and deliver tailored content recommendations.

The project will focus on:

- **Multi-source aggregation**: Collecting articles from trusted sources.
- **Adaptive simplification**: AI-generated rewrites tailored to a reader’s comprehension level (i.e. beginner, intermediate, advanced).
- **Contextual help**: Popups with definitions, explanations, and references.
- **Accessibility support**: Screen reader compatibility, adjustable text, high-contrast modes.
- **Personalization**: User preferences for categories, saved articles, and adjustable complexity.
- **Recommendations**: Learning from user behavior to suggest relevant content and new topics to explore.

This ensures a problem-solution fit by addressing both comprehension and usability challenges, while creating a personalized, inclusive reading experience.

Literature Review on Existing and Similar Solutions

1) General News Aggregators

- **Examples**: Google News, Apple News
- Provide personalized feeds based on browsing history and interests but assume prior knowledge.
- Articles are delivered as-is, without simplification, making them difficult for readers unfamiliar with a topic.

2) AI News Applications

- **Examples**: Rize
- Utilizes AI to summarize articles into two concise paragraphs, reducing length but not necessarily adapting complexity.
- Summaries may omit important context, and there is no personalization by knowledge level.

3) Specialized News Applications

- **Examples**: Flipboard, Ground News
- Platforms that emphasize unique perspectives on news consumption.
- Flipboard allows users to curate a "magazine" of interests, focusing on discovery and personalization through curation, while Ground News compares coverage across outlets to highlight bias and differences in framing.
- Both provide alternative lenses for engaging with news but do not address the comprehension barrier caused by jargon or assumed prior knowledge.

4) Educational Platforms

- **Examples**: Newsela, The Skimm
- Simplify content for readers, often rewriting at different grade levels (Newsela) or providing condensed daily updates (The Skimm).
- However, these are designed for niche audiences (K–12 or casual readers) rather than general-purpose news consumption.

5) Educational Video Platforms

- **Examples**: Ted-Ed, YouTube, Khan Academy
- Provide accessible explanations of complex topics in multimedia formats.
- While effective for deep learning, these platforms are not tied to real-time news and do not offer continuous updates.

6) Accessibility Browser Extensions

- **Examples**: Read Aloud, Helperbird
- Provide single-function support such as text-to-speech, font resizing, or contrast adjustments.

- They improve usability but are external add-ons, lacking integration with content simplification or personalization features.

7) General-Purpose LLMs

- **Examples**: Grok, Perplexity, ChatGPT, Deepseek
- Offer on-demand explanations and the ability to answer user queries about difficult material.
- However, they require active prompting and are not optimized for continuous, article-based reading experiences.

Gap: No platform integrates simplification, personalization, and accessibility-first design into a unified, user-friendly news experience.

Product Vision Statement

For news readers of all backgrounds, including beginners and individuals with accessibility needs, the News for U platform is an AI-powered, accessible news aggregator that simplifies complex news content, explains jargon, and adapts articles to each reader's level of understanding.

Unlike traditional news aggregators that assume expertise or accessibility extensions that address only technical barriers, our product integrates AI-driven simplification, comprehension-based personalization, and built-in accessibility to create a more inclusive and approachable news experience.

Target Audience and Stakeholders

1) Primary Users

- **Novice news readers**: Students, young adults, or lifelong learners exploring unfamiliar topics (i.e. finance, politics, sports, health, etc.).
- **Accessibility-dependent users**: Individuals with visual, auditory, or cognitive disabilities who rely on screen readers, high-contrast modes, or adjustable text.
- **Time-constrained readers**: Professionals or casual readers who want quick comprehension without sacrificing accuracy.

2) Secondary Stakeholders

- **Educators and trainers**: Using simplified news for teaching media literacy or domain-specific knowledge.
- **Institutions promoting accessibility**: Libraries, nonprofits, or companies ensuring inclusive digital experiences.
- **Content providers and aggregators**: Media outlets interested in reaching broader audiences or experimenting with AI-assisted content delivery.

Success Criteria

- **User Comprehension**: ≥80% of users report improved understanding of previously unfamiliar articles.
- **Accessibility Impact**: Platform meets Web Content Accessibility Guidelines (WCAG) 2.1 AA Compliance.
- **Engagement**: Users spend ≥25% more time per session on the platform compared to reading unmodified articles.
- **Simplification Accuracy**: ≥80% of AI-simplified articles preserve the original meaning, verified via automated semantic similarity checks and human review for a sample set.
- **Performance**: Median simplification time ≤15 seconds for articles with <1,500 words.

Risk Assessment

1) Technical Risks

- **AI Misinterpretation and Bias:** Simplified content could unintentionally mislead or introduce bias.
    - **Mitigation:** Implement human-in-the-loop review for initial batches; use explainable AI metrics to monitor semantic fidelity.
- **Performance Bottlenecks:** High traffic or large articles may slow down the system.
    - **Mitigation:** Use asynchronous processing, caching, and scalable cloud services (i.e. AWS Lambda and S3).
- **Data Privacy and Security:** User preferences and reading history may contain sensitive data.
    - **Mitigation:** Encrypt data in transit and at rest.
- **LLM API Limitations:** Dependence on LLMs may restrict usage.
    - **Mitigation:** Cache articles, respect rate limits, and implement fallback mechanisms.

2) Organizational Risks

- **Content Licensing**: Dependence on third-party news sources and APIs introduces potential legal and contractual risks.
    - **Mitigation**: Prioritize using open-source news providers, and maintain a curated list of legally compliant news providers.
- **User Adoption Challenge:** Users may default to familiar news platforms.
    - **Mitigation:** Deploy pilot studies in schools or universities to help highlight the platform's unique combination of simplification, accessibility, and personalization.
- **Scope Creep:** Expanding AI features (i.e. multimedia summaries, knowledge graphs) may delay MVP.
    - **Mitigation:** Define strict MVP features for first release and use Agile sprints to prioritize features.